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Highlights

- The effect of electron affinity of ZnOS used as a buffer layer in MoS₂-based solar cells was investigated and ZnOS buffer layer significantly enhanced the photovoltaic performance of the MoS₂ absorber layer.
- Through an integrated optimization methodology, the power conversion efficiency (PCE) was improved from 29.17% to 31.42%.
- The presented approach offers higher efficiency gains, lower computational cost, statistically validated parameter interactions, and a repeatable and scalable optimization framework.

Integrated SCAPS-1D Modeling and Evolutionary Optimization for High-Performance MoS₂/ZnOS Heterojunction Solar Cells

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Abstract

This study presents a systematic optimization strategy for an environmentally friendly Au/Sb₂S₃/MoS₂/ZnOS/FTO thin-film solar cell by integrating SCAPS-1D numerical simulation with advanced statistical and computational methods, including Design of Experiments (DOE), Response Surface Methodology (RSM), Box–Behnken Design (BBD), and Genetic Algorithm (GA) optimization. Unlike conventional one-factor-at-a-time approaches, multiple key device parameters such as MoS₂ electron affinity, absorber thickness, doping concentration, defect density, and ZnOS electron affinity were simultaneously optimized to maximize power conversion efficiency (PCE). As a result, the PCE increased from 29.170 % to 31.416 %, while re-evaluation of the GA-optimized structure in SCAPS-1D yielded a closely matched efficiency of 31.420 %, confirming the reliability of the optimization framework.

GA-based sensitivity analysis indicated that defect density and interfacial band alignment are the most influential parameters controlling open-circuit voltage and overall device performance. Beyond improving the specific Sb₂S₃-based architecture, the proposed methodology demonstrates an efficient design-space engineering approach that enhances predictive accuracy while reducing computational cost. Therefore, this integrated framework offers a scalable and cost-effective route for the rational design and optimization of MoS₂-based and other transition metal dichalcogenide photovoltaic systems, contributing to the development of next-generation high-efficiency chalcogenide solar cells.

Keywords: MoS₂; ZnOS; Solar cell; SCAPS-1D; Integrated methodology.

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1. Introduction

Thin film solar cells (TFSCs) have become an important research topic in photovoltaic (PV) technologies due to their large-scale production capacity, cost effective fabrication processes, and high *PCE* achievable compared to traditional silicon (Si) based solar cells [1]. Recently, among commercially available TFSC structures, organic, perovskite, polycrystalline CdTe, CIS, and CIGS- based absorber layers remained popular [2, 3]. Over the past decade, extensive experimental and theoretical investigations of these structures have led to significant cost reductions, with *PCE* values reaching 23.64 % experimentally and 31.15 % in theoretical predictions [4, 5]. However, currently commercialized technologies remain constrained by significant limitations. In particular, the presence of cadmium (Cd), a toxic element in these structures, raises serious environmental concerns [6]. Furthermore, production costs represent a major obstacle to high production capacity [7-9]. These limitations have driven researchers to focus on materials composed of non-toxic elements, earth-abundant with the objective of developing low-cost, high-efficiency next-generation solar cells [10,11]. In this context, all alternative potential photovoltaic materials are expected to possess these fundamental properties. Conducting comprehensive and systematic research on these properties is of great importance to the scientific community. Among the currently available commercial solar cell technologies, including amorphous silicon (a-Si), dye-sensitized solar cells (DSSCs), organic PV, and perovskite solar cells, transition metal dichalcogenides (TMDs) are attracting significant attention due to their high absorption coefficients, nearly ideal bandgaps, and self-passivating surfaces [12, 13]. Additionally, TMD thin films have been extensively studied in the literature in recent years within the framework of alternative absorber layer materials [13].

TMDs are generally represented by the formula MX_2 , where M denotes a transition metal (Mo, Nb, W, Ti, Re, Ta) and X represents a chalcogen element (S, Se, Te). Owing to their diverse crystal phases, these compounds enable the formation of robust heterostructures that minimize lattice mismatch and exhibit superior interfacial properties [14]. Among TMDs, molybdenum disulfide (MoS_2) has offers significant advantages in photovoltaic device applications due to its flexible bandgap ranging from 1.2 eV to 1.9 eV, n-type and/or p-type electrical conductivity, superior carrier mobility, high absorption coefficient, high optical transparency, and large surface area [2, 15-20]. MoS_2 naturally exhibits n-type conductivity due to the presence of sulfur vacancies; however, it can be converted to p-type conductivity through doping with P, O, or Nb. A p- MoS_2 layer with high electrical quality preserves proper band

alignment at the back contact and enhances carrier separation, which contributes to an increase in the open-circuit voltage (V_{oc}) in chalcogenide thin film solar cells [21]. Due to these properties, p-type MoS₂ is also preferred in theoretical device simulations related to solar cells.

The triangular prismatic and octahedral structural properties of MoS₂ enable the error-free occurrence of photogeneration, primarily originating from the d-orbital [15]. The antibonding properties of the conduction and valence bands of MoS₂ leads to the emergence of photocorrosion resistance, which provides improved stability for solar cell applications [17, 18]. Compared to graphene, which is widely used in photovoltaic applications, MoS₂ has numerous advantages, particularly in terms of its abundance in nature, low cost, and high visible light absorption capacity [22]. MoS₂ has become a popular material in photovoltaic applications due to these properties [23]. For example, it has been experimentally proven that adding MoS₂ as a hole transport layer (HTL) to flexible organic polymer solar cells increases efficiency by 3.56 % [24].

Over the past five years, experimental research on MoS₂ in solar cells has primarily focused on improving device performance [25-28], device hybridization [29, 30], synthesis routes [31-33], transfer methods [34, 35], mechanical properties and structural integrity [36, 37] and evaluation of MoS₂'s intrinsic properties [38-40] and device performance has been reported to reach a maximum efficiency of 22.5 % [41]. On the other hand, there are very few studies in the literature on the theoretical investigation of MoS₂ [2, 42-48]. The limited number of theoretical studies conducted to date and the fact that the highest reported theoretical efficiency remains around 26 % highlight the need for a more comprehensive investigation of MoS₂, which holds great potential as an absorbent material. Although these findings suggest that TMDC-based heterojunction structures could theoretically achieve efficiencies of up to ~33.65 % according to the Shockley–Queisser limit, theoretical studies in this area remain insufficient [49, 50]. According to the Shockley–Queisser limit, the maximum theoretical efficiency for single-junction solar cells is approximately 33% at an optimal band gap of ~1.3-1.4 eV. The band gap of MoS₂ (~1.29-1.50 eV) lies very close to this optimal range, indicating its strong potential for high theoretical efficiency when combined with appropriate interface engineering and minimized loss mechanisms. However, it should be emphasized that the reported maximum *PCE* corresponds to a simulation-based upper-limit efficiency obtained under optimized material and interface conditions, rather than an experimentally realized device performance [51].

In general, theoretical studies in this field commonly employ numerical simulation software packages such as the One-Dimensional Solar Cell Capacitance Simulator (SCAPS-1D),

Automatic Forward Bias Simulation of Heterostructures with External Terminals (AFORS-HET), and Analysis of Microelectronic and Photonic Structures-1D (AMPS-1D) [2, 40-47]. Among these, SCAPS-1D was utilized in the present study.

To the best of our knowledge, statistical optimization approaches such as DOE, RSM, BBD, and GA implemented through the Matrix Laboratory (MATLAB) optimization toolbox for training and testing have not yet been applied in the literature to efficiently optimize the complex parameters of MoS₂ solar cells, such as film thickness, doping concentration, electron affinity, defect density, and different buffer layers, or to reduce the effects of production variability. The absence of advanced statistical optimization methods in existing theoretical studies involving MoS₂ limits the comprehensive exploration of the performance potential of MoS₂-based solar cells. Furthermore, it reduces the effectiveness of the device optimization process by making it difficult to systematically and rapidly identify fundamental design constraints. To the best of our knowledge, no previous study has combined all these statistical approaches for the optimization of the Au/Sb₂S₃/MoS₂/ZnOS/FTO solar cell structure.

In this study, the Au/Sb₂S₃/MoS₂/ZnOS/FTO solar cell structure was theoretically investigated. While existing studies largely rely on parametric sweeps and limited variable analyses, this research enables a systematic and high-accuracy exploration of the multi-parameter device design space by integrating SCAPS-1D numerical device simulations with advanced statistical optimization approaches based on DOE, RSM, BBD, and GA. Therefore, this study provides an original contribution by filling a significant methodological gap in the literature regarding the theoretical design of MoS₂-based thin-film solar cells.

In this respect, the study not only determines the maximum-efficiency point by quantitatively analyzing the interactions among MoS₂ thickness, doping concentration, electron affinity, defect density, and buffer layer parameters of antimony(III) sulfide (Sb₂S₃), but also identifies the fundamental design constraints and sensitivity regions. Thus, the reliability of device performance predictions is improved, and a cost effective and time efficient design framework is provided to guide experimental studies.

The integration of multiple statistical optimization techniques into MoS₂-based photovoltaic structures for the first time transforms the study into not only a material focused innovation but also a methodology driven one, thereby introducing a high impact approach to the literature for the rational design of next generation chalcogenide based solar cells.

2. Theoretical modeling and optimization methodology

2.1. Theoretical modeling

The numerical analysis of the Au/Sb₂S₃/MoS₂/ZnOS/FTO (SMZF) solar cell structure was performed using SCAPS-1D, developed by Prof. Dr. Marc Burgelman and his research group at Ghent University [52, 53]. SCAPS-1D is a simulation software widely used in the literature for analyzing the electrical and optoelectronic properties of homo and heterojunction solar cells [54-57]. Furthermore, numerous studies in the literature validate the reliability of the SCAPS-1D model by comparing simulation outcomes with experimental data. For instance, a study on monocrystalline silicon solar cells reported that results obtained from SCAPS-1D exhibit a high degree of agreement with corresponding experimental measurements, thereby confirming the model's predictive accuracy [58].

In this study, the SMZF solar cell structure was numerically analyzed using SCAPS-1D, and DOE, RSM, BBD, and GA approaches were integrated to overcome the limitations of the software in evaluating multi-parameter interactions. To the best of our knowledge, no study has yet been reported in the literature in which these methods have been jointly applied to MoS₂-based solar cells. Therefore, the proposed approach is methodologically unique. This integrated optimization framework overcomes the local optimization and accuracy limitations of conventional SCAPS-1D studies by enabling the systematic, rapid, and reliable determination of design parameters.

2.2. Numerical analysis of the SMZF solar cell using SCAPS-1D

SCAPS-1D is a numerical simulation software developed specifically to examine the optical and electronic properties of thin film solar cells, and is widely used in both academic and industrial fields [53]. This software is designed to simulate the performance of one-dimensional semiconductor devices with multilayer heterostructures. SCAPS-1D can generate a wide range of characterization outputs based on user-defined physical parameters, including band gap, electron affinity, dielectric constant, defect density, optical absorption coefficient, doping concentration, layer thickness, and operating temperature.

In SCAPS-1D simulations, Shockley-Read-Hall (SRH), radiative, and Auger recombination mechanisms can be explicitly defined [59]. In this study, SRH recombination via defect states and interface traps is assumed to be the dominant recombination mechanism in the polycrystalline $\text{Sb}_2\text{S}_3/\text{MoS}_2/\text{ZnOS}$ heterojunction structure [60]. This assumption is primarily justified by the presence of grain boundaries and high defect densities in thin-film materials, which significantly limit carrier lifetime. Radiative recombination is considered a secondary process, while Auger recombination is assumed to be negligible under moderate doping levels and standard one-sun illumination conditions [61]. Consequently, the optimization analysis particularly emphasizes the critical influence of defect density on overall device performance.

SCAPS-1D works by solving the fundamental equations governing the behavior of semiconductor devices. These equations include the Poisson equation and the continuity equations and enable the analysis of the steady-state performance and capacitance-voltage ($C - V$) characteristics of a solar cell. The fundamental physical equations governing the device performance are given as follows:

- a. The Poisson equation relates the electrostatic potential (V) of a semiconductor to the spatial charge density (ρ) (Eq. (1)) [1,11].

$$\frac{d}{dx} \left(\epsilon \frac{dV}{dx} \right) = -\rho \quad (1)$$

- b. The open-circuit voltage (V_{oc}), PCE , current-voltage ($J - V$) characteristics, short-circuit current density (J_{sc}), fill factor (FF), and other performance parameters are calculated using the electron and hole continuity equations. It relates the conservation and flow of electron (n) and hole (p) current densities (J_n , J_p) to the generation (G) and recombination (R) rates. These equations are equal to zero under steady-state conditions (Eqs. (2), (3)) [1,11].

$$\frac{1}{q} \frac{dJ_n}{dx} = G - R_n = 0 \quad (2)$$

$$-\frac{1}{q} \frac{dJ_p}{dx} = G - R_p = 0 \quad (3)$$

The Eqs. (4)-(6) used to calculate the basic parameters of solar cells are given below [1].

$$\nabla \cdot \varepsilon \nabla \varphi = -q(p - n + N_D - N_A) \quad (4)$$

$$\nabla \cdot J_n = -q(R - G) + q \frac{\partial n}{\partial t} \quad (5)$$

$$-\nabla \cdot J_p = -q(R - G) + q \frac{\partial p}{\partial t} \quad (6)$$

Here, q represents the fundamental charge, G represents the photo-production rate, and R represents the total recombination rate, which includes the Shockley-Read-Hall (SRH), radiation (band-to-band), and Auger recombination mechanisms [1]. Additionally, n and p are the concentrations of free carriers of electrons and holes, q is the electronic charge, φ is the electric potential, and ε is the dielectric constant of the material in the equation. N_D and N_A are the concentrations of ionizing donors and acceptors respectively, J_n and J_p are the electron and hole current density, respectively. G and R are the generation rate and electron-hole recombination rate, respectively. The FF can be calculated from Eq. (7) as follows [1, 62, 63];

$$FF = \frac{P_{max}}{P_{in}} = \frac{V_{max} \cdot I_{max}}{V_{oc} \cdot J_{sc}} \quad (7)$$

The percentage of energy conversion efficiency (PCE) is given by Eq. (8).

$$PCE (\%) = \frac{P_{max}}{P_{in}} = \frac{V_{oc} \cdot J_{sc} \cdot FF}{P_{in}} \quad (8)$$

$$V_{oc} = \frac{KT}{q} \ln \left[\frac{J_{sc}}{J_o} + 1 \right] \quad (9)$$

The V_{oc} is related to both the J_{sc} and the J_o , and also depends on the thickness of the absorber layer (Eq. (9)). The schematic representation of the SMZF solar cell structure used in this study is shown in Fig. (1). In the SMZF solar cell architecture, fluorine-doped tin oxide (FTO) coated glass, ZnOS, p-type MoS₂, p-type Sb₂S₃ and Au were used as the front contact, the buffer layer, the absorber layer, the back surface field (BSF) layer and the back contact, respectively.

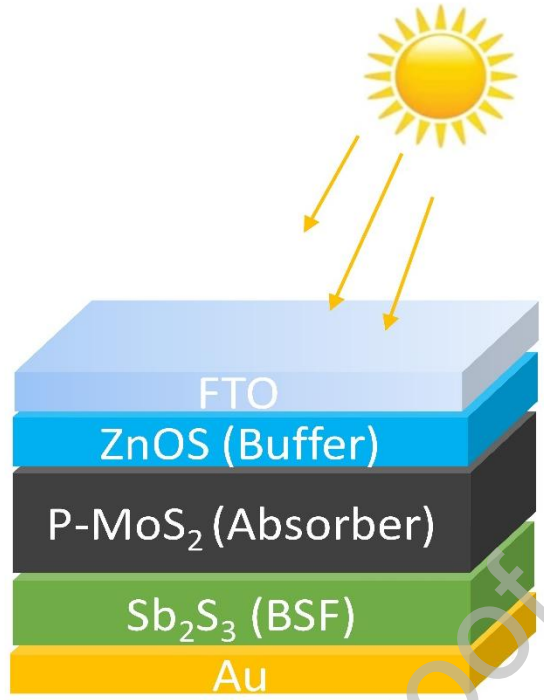


Fig. 1. The device configuration of the SMZF solar cell heterostructure.

The defect parameters of absorber and physical device parameters used in the SCAPS-1D simulations are presented in Tables 1 and 2.

Table 1.

Defect Parameters of Absorber [43, 44].

Parameters	<i>p</i> -MoS ₂
Defect density, N_t (cm ⁻³)	Varied
Defect type	Neutral
Capture cross-section electrons (cm ²)	1.000E+15
Capture cross-section holes (cm ²)	1.000E+15
Energetic distribution	Single
Reference for defect energy level, E_t	Above E_v
Energy level concerning reference (eV)	0.6

The standard AM1.5G spectrum, corresponding to an irradiation intensity of 1000 W/m^2 and a temperature of 300 K , was used to model the solar spectrum. This spectrum is widely accepted as a reference for performance evaluation in photovoltaic device simulations [64]. The SCAPS-1D software provides a powerful and reliable simulation platform, particularly for the optimization of next generation thin film absorber materials such as MoS₂ and for the systematic investigation of defect effects at heterojunction interfaces [1, 64].

The parameter ranges used in the optimization study were determined based on experimentally reported values in the literature for $\text{Sb}_2\text{S}_3/\text{MoS}_2/\text{ZnOS}$ heterojunction solar cells, as well as physically realistic operating conditions [65]. Defect density ranges were selected to represent typical values observed in polycrystalline thin films, while doping concentrations were constrained within practically achievable levels. Electron affinity values were chosen in accordance with reported literature data for the respective materials and were varied within narrow intervals to investigate the influence of interfacial band alignment. Layer thicknesses were defined by considering values attainable through thin-film deposition techniques, ensuring an appropriate balance between optical absorption and carrier transport. Accordingly, all optimization variables were defined within physically meaningful and experimentally justified boundaries rather than arbitrary ranges [65, 66].

Table 2.

Physical parameters and values used in the simulation.

Parameter and Units	Sb_2S_3	MoS_2	ZnOS	FTO
Layer type	BSF	Absorber	Buffer	Window
Conductivity type	p^+	p	n	n^+
Thickness (nm)	10	Varied	50	50
Energy gap, E_g (eV)	1.70	1.29	3.10	3.60
Electron affinity, χ (eV)	3.8	Varied	Varied	4.00
Dielectric permittivity, ϵ_r	7.0	9.00	9.0	9.00
Effective conduction band density of states, N_C (cm^{-3})	$2.500\text{E}+19$	$5.190\text{E}+17$	$2.200\text{E}+18$	$2.200\text{E}+18$
Effective valence band density of states of states, N_V (cm^{-3})	$3.500\text{E}+19$	$7.320\text{E}+18$	$1.800\text{E}+19$	$1.700\text{E}+19$
Electron thermal velocity (cm/s)	$1.000\text{E}+7$	$1.000\text{E}+7$	$1.000\text{E}+7$	$1.000\text{E}+7$
Hole thermal velocity (cm/s)	$1.000\text{E}+7$	$1.000\text{E}+7$	$1.000\text{E}+7$	$1.000\text{E}+7$
Electron mobility, μ_n ($\text{cm}^2/\text{V/s}$)	$7.00\text{E}-2$	$2.000\text{E}+2$	$1.00\text{E}+2$	$1.00\text{E}+2$
Hole mobility, μ_p ($\text{cm}^2/\text{V/s}$)	$2.00\text{E}-2$	$8.500\text{E}+0$	$2.500\text{E}+1$	$2.50\text{E}+1$
Acceptor concentration, N_A (cm^{-3})	$1.000\text{E}+18$	Varied	-	-
Donor concentration, N_D (cm^{-3})	-	-	$2.00\text{E}+18$	$1.00\text{E}+18$
Temperature, (K)		300		
Reference	[67]	[44,45,67,68]	[69-71]	[72,73]

2.3. Practical Applicability and Experimental Perspective

In this study, the electron affinity of the ZnOS (Zn(O,S)) buffer layer is considered experimentally tunable through variation of the O/S ratio, consistent with reported fabrication techniques such as reactive sputtering, chemical bath deposition (CBD), and atomic layer deposition (ALD). Accordingly, the parameter range used in this work agrees well with values

reported in the literature. The defect density value on the order of $\sim 10^{12} \text{ cm}^{-3}$ is regarded as an idealized lower bound; however, recent advances in deposition processes, post-deposition treatments, and interface passivation techniques suggest that such low defect densities can be approached in high-quality thin-film structures. Furthermore, interfacial band alignment, identified as a key factor in the optimization process, can be effectively controlled through buffer layer engineering and surface modification strategies. Literature reports further confirm that the electronic properties of Zn(O,S) and defect states in Sb_2S_3 -based heterojunctions can be systematically tuned via composition control and fabrication conditions [74-77]. Overall, these findings support the physical validity and experimental feasibility of the parameter ranges adopted in this study.

2.4. Model Limitations

This study is based on SCAPS-1D numerical simulations, which inherently involve certain limitations. First, SCAPS-1D employs a one-dimensional (1D) modeling approach and does not account for two- or three-dimensional effects. Consequently, lateral losses that may arise in large-area devices or structures with non-uniform morphology are not explicitly considered. Second, the model does not explicitly represent grain boundaries. Although grain boundaries play a significant role in carrier recombination in thin-film solar cells, SCAPS-1D incorporates their effects only indirectly through effective defect density parameters, rather than modeling them as distinct physical entities. Third, interfacial physics is treated in an idealized manner. Complex phenomena such as interface states, Fermi-level pinning, and chemical bonding effects are simplified using boundary defect states and fixed parameter assumptions. Despite these limitations, SCAPS-1D remains a widely validated and extensively used simulation tool in the field of thin-film photovoltaics, as it successfully captures key physical processes such as carrier transport, recombination mechanisms, and band alignment effects [60, 78]. In addition, the integration of Design of Experiments (DOE), Response Surface Methodology (RSM), and genetic algorithm-based optimization enables a systematic exploration of the multi-parameter design space, despite the deterministic nature of the underlying model. Therefore, rather than providing an exact prediction of absolute device performance, this study primarily aims to identify the relative influence of design parameters and to reveal optimization trends within the device architecture.

2.5. Optimization methodology

In this study, a comprehensive theoretical modeling framework was developed by integrating DOE, RSM, BBD, and GA approaches. This methodology combines the deterministic semiconductor device simulations performed using SCAPS-1D with advanced statistical modeling and evolutionary optimization techniques. Accordingly, the study goes beyond conventional single-parameter analyses and offers a systematic and multi-dimensional optimization strategy.

2.5.1. Design of Experiments (DOE)

The DOE is a statistical methodology that allows systematic and balanced variation of selected variables within defined ranges in experimental or numerical simulation environments, influencing key system performance parameters (such as layer thickness, doping concentration, electron affinity, defect density). In this study, DOE was employed as a systematic approach to efficiently sample the high-dimensional parameter space in terms of information content, rather than relying on random or exhaustive search methods. Consequently, the number of required simulations is minimized while still enabling the accurate identification of both main effects and interaction effects among the parameters.

2.5.2. Response surface methodology (RMS)

The RSM can represent the relationship between device outputs (PCE , V_{oc} , J_{sc} and FF) and input parameters using simulation results obtained with the DOE through second-degree polynomial models by Equation 10.

$$Y = \beta_0 + \sum \beta_i X_i + \sum \beta_{ii} X_i^2 + \sum \beta_{ij} X_i X_j \quad (10)$$

Here, Y and X_i represents the PCE and the design parameters, respectively. β_0 is constant and β_i is coefficient linear effect of the design parameters. This model enables the mathematical definition of nonlinear interactions between parameters and the expression of performance as a continuous surface [78, 79].

2.5.3. Box–Behnken Design (BBD)

The BBD is one of the second-order modeling designs used for RSM and it is particularly useful for providing highly accurate predictions while reducing the number of experiments in three-level factor structures [80]. Factor combinations are selected at the edge centers of the design space, thereby preventing physically unrealistic simulations at extreme values. In this study, BBD was employed to establish an optimum sampling plan for the SCAPS-1D simulations and to develop a reliable response surface model.

2.5.4. Genetic Algorithm (GA)

GA is a stochastic optimization method based on the principles of natural selection and genetic evolution. Each individual was defined as a chromosome representing the design parameters of the solar cell. *PCE* was selected as the fitness function, and this selection produced new generations through crossover and mutation operations. The greatest advantage of GA is that it can find solutions close to the global optimum in a multimodal design space without getting stuck at local maxima [81, 82]. GA has been used to understand the simultaneous interactions between variables. Ultimately, SCAPS-1D provides physics-based device simulation, while the DOE-RSM-BBD combination represents parameter interactions through a statistically modeled response surface in this integrated methodology. GA performs global optimization on this continuous model.

3. Results and discussion

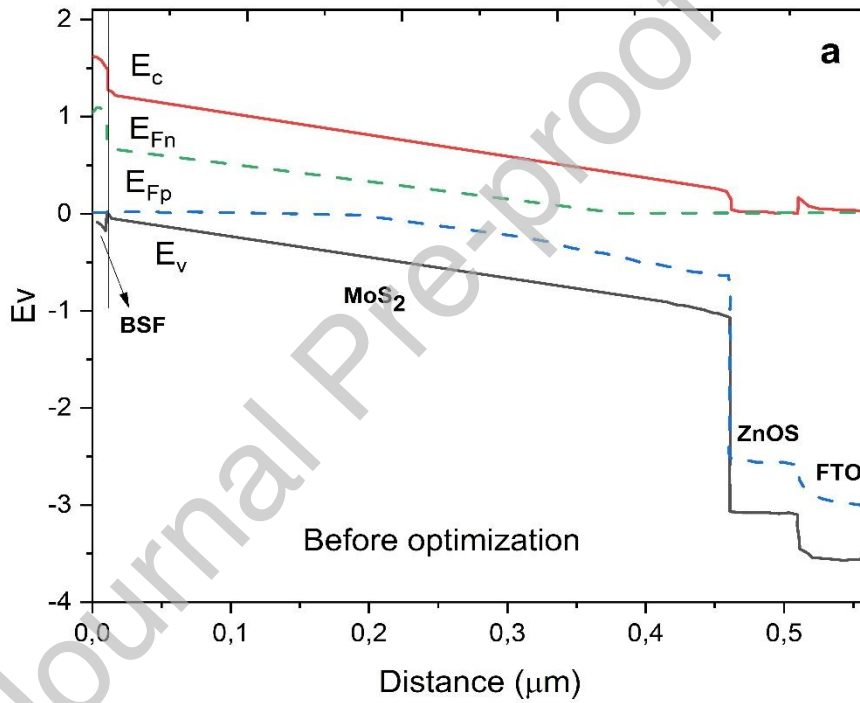
Traditional OFAT methods, commonly found in the literature, change only one parameter while keeping other parameters constant and neglecting parameter interactions; in contrast, SCAPS-1D simulation integrated with DOE and GA allows for the simultaneous examination of the effects of all parameters. This hybrid framework enables the simultaneous evaluation of the effects associated with multiple parameters. Thus, it facilitates the global exploration of the design space and leads to a more robust, physically meaningful, and statistically reliable optimization methodology.

To the best of our knowledge, this study is among the few that integrate DOE, RSM, and GA with SCAPS-1D for the systematic optimization of MoS₂-based solar cells. In contrast to studies based on the OFAT strategy, the proposed methodology simultaneously accounts for

parameter interactions. This approach provides a robust, physically consistent, and statistically reliable optimization strategy and represents a methodological advancement beyond traditional simulation-based device studies.

3.1. Band Alignment Optimization and Its Impact on Device Performance

The electron affinities (χ_e) of MoS₂ and ZnOS determine the conduction band offset between MoS₂ and ZnOS. The energy band alignment at the p-MoS₂/n-ZnOS interface plays a crucial role in charge separation, interfacial recombination, and overall photovoltaic performance [83].



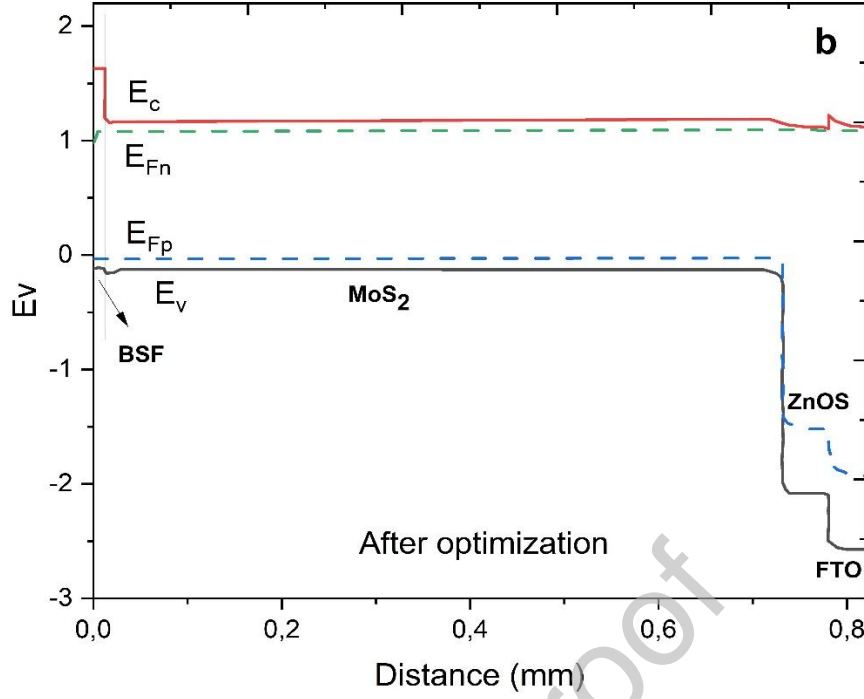


Fig. 2. Energy band diagram of the SMZF solar cell a) before optimization and b) after optimization.

Fig. 2 (a, b) shows the simulated energy band diagrams of the p-MoS₂/ZnOS heterojunction before and after band alignment optimization, respectively. Figure 2 (a) shows a distinct band bending within the absorber and at the p-MoS₂/ZnOS interface, indicating non-ideal band alignment. The discontinuity in the interface creates a distinct step-like band offset that promotes electron-hole recombination at the interface. This unfavorable alignment leads to increased recombination losses, manifested as a decrease in open-circuit voltage and a reduction in fill factor [85]. Following optimization, Fig. 2(b) shows that the interface band alignment has been significantly improved.

By shifting the conduction band offset to a small negative value ($\Delta E_C \approx -0.050$ eV to -0.098 eV) (Table 3), an optimal band configuration is achieved, which facilitates efficient electron transport while suppressing excessive interfacial recombination [85, 86]. The Anderson model, also known as the electron affinity rule (EAR), predicts band alignment in ideal bulk semiconductor heterojunctions [86]. The conduction band offset (ΔE_C) and valence band offset (ΔE_V) were calculated using the Anderson model (Eqs. (11) and (12)) [87]. The conduction and valence band offset values are given in Table 1.

$$\Delta E_C^{Anderson} = \chi_{ZnOS} - \chi_{MoS_2} \quad (11)$$

$$\Delta E_V^{Anderson} = (E_{g,ZnOS} - E_{g,MoS_2}) + \Delta E_C^{Anderson} \quad (12)$$

Considering the band alignment of the two materials, semiconductor heterostructures can be classified into three types: type-I, type-II, and type-III. The band alignment obtained in this study corresponds to a type-II heterojunction. After optimization, a reduction in the cliff type is observed in the energy band diagram. This suppresses interfacial recombination while maintaining efficient electron transport, leading to improved device performance [85, 88].

Band alignment, carrier selectivity, and interface recombination are among the key factors determining the performance of heterojunction solar cells. According to the Anderson electron affinity model, the conduction band offsets (ΔE_C) at the Sb_2S_3/MoS_2 and $MoS_2/ZnOS$ interfaces were calculated using the electron affinities of the constituent materials. Considering electron affinity values of approximately 3.9 eV for Sb_2S_3 , 4.1 eV for MoS_2 , and 3.8-4.0 eV for ZnOS, a small positive spike-type alignment is formed at the Sb_2S_3/MoS_2 interface [90]. This configuration promotes selective electron transport while suppressing backflow of holes, thereby reducing interface recombination. It has been reported in the literature that positive conduction band offsets in the range of 0-0.3 eV are optimal for high-efficiency thin-film solar cells. As a result of the optimized parameters, an appropriate band alignment was achieved, leading to an improvement in V_{OC} , fill factor (FF), and overall power conversion efficiency [91].

Table 3.

Theoretical band offset values obtained using the Anderson model.

Materials	E_g (eV)	Affinity (eV)		$\Delta E_C^{Anderson}$		$\Delta E_V^{Anderson}$	
		Before Opt.	After Opt.	Before Opt.	After Opt.	Before Opt.	After Opt.
p-MoS ₂	1.29	4.250	4.250	-0.050	-0.098	1.760	1.712
n-ZnOS	3.10	4.200	4.152				

Compared to other buffer layers such as TiO_2 , ZnO, CdS and ZnOS enables efficient electron extraction while suppressing recombination by providing near-optimal band alignment with a small shift in the conduction band [91, 92]. This unique balance explains the superior performance of the ZnOS-based device architecture. Furthermore, ZnOS also allows tuning of electron affinity by adjusting the S/O ratio [91]. This provides a realistic way to adjust the band alignment without altering the intrinsic properties of the absorber.

The consistency between the optimized band diagram and the band shifts predicted by the Anderson model confirms the physical validity of the adopted approach (Table 3). In

general, the optimization strategy provides an optimum balance among optimized p-MoS₂/ZnOS band alignment, charge transport, and recombination suppression, explaining the superior photovoltaic performance achieved after optimization.

3.2. Device performance

In this study, the affinity of ZnOS and the affinity, thickness, N_t and N_a parameters of MoS₂ were systematically investigated for the first time for the SMZF solar cell heterojunction structure. The J - V and QE graphs for this structure obtained before and after optimization are shown in Fig. 3 and Fig. 4, respectively. V_{oc} , J_{sc} , FF and PCE values were observed to increase after optimization. These values were calculated as $V_{oc} = 1.04$ V, $J_{sc} = 34.302$ mA/cm², $FF = 88.16$ %, and $PCE = 31.420$ %.

Table 4.

Compression between optimized process parameters by GA and SCAPS-1D.

Before Optimization	After Optimization		
PCE (%) BBD	PCE (%) GA	PCE (%) SCAPS-1D	Error (%)
29.170	31.416	31.420	0.013

The highest PCE value predicted by the regression equation obtained after the BBD analysis of the RSM prior to optimization was calculated as 29.170 %. When this regression equation was run using MATLAB's GA optimization tool, $PCE = 31.416$ % was calculated corresponding to the optimum values of the affinity, thickness, N_t , and N_a parameters. SCAPS-1D was rerun to test these values, and $PCE = 31.420$ % was calculated with an error of 0.013 % (Table 4).

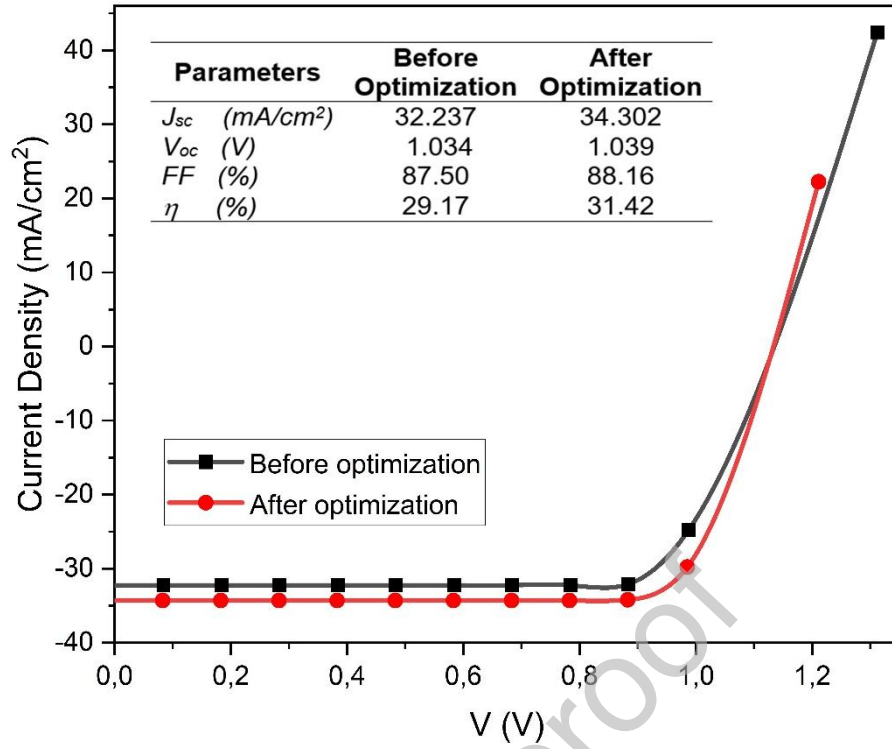


Fig. 3. $J - V$ graph of the before and after optimization in SMZF solar cell structure.

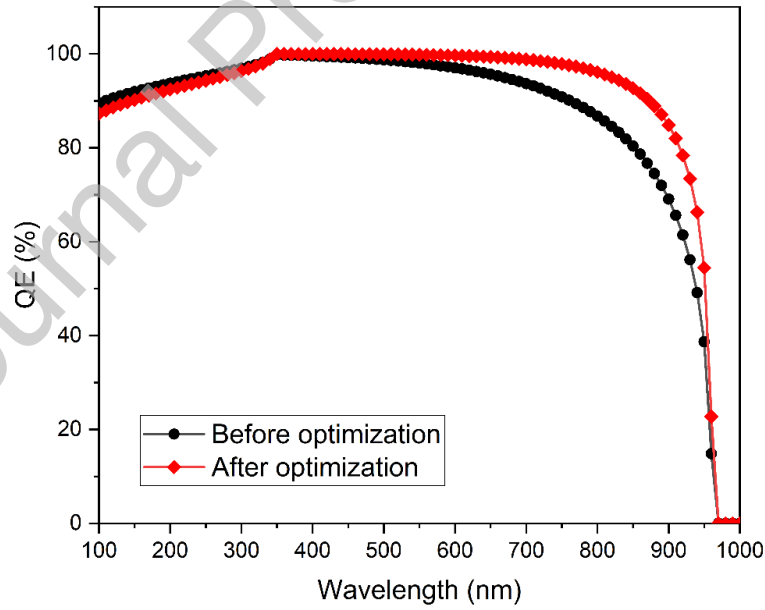


Fig. 4. QE graph of the before and after optimization in SMZF solar cell structure.

It was observed that the current density increased from $\sim 32 \text{ mA}/\text{cm}^2$ to $\sim 34 \text{ mA}/\text{cm}^2$ after optimization in the $J - V$ graph (Fig. 3). This is consistent with the QE graph in the near-infrared regions shown in Fig. 4. The increase in QE between 600 nm and 1000 nm shows that surface and volumetric recombination are simultaneously reduced, as well as carrier collection

efficiency is improved. Furthermore, it confirms that optimization effectively reduces defect-assisted recombination pathways [93]. These results demonstrate that the optimization strategy leads to a significant increase in overall photovoltaic performance.

3.3. Response surface methodology (RSM)

RSM, developed by Box and Wilson in the 1950s, is used in tools such as Statistica and Design Expert [78]. This approach enables the creation of an empirical model and optimization by integrating experimental design with statistical analysis. Statistica is used in various tasks such as testing, characterization, adaptive parameter design, comparison testing, and optimization. In this study, BBD, one of the RSM methods, was used to optimize the photovoltaic performance of the SMZF solar cell structure.

3.3.1. Design Matrix Development

In this study, the low and high points of the input parameters to be used in the RSM-BBD analysis were determined by considering the literature (Table 5).

Table 5.

Input parameters

Parameters	Units	Low	High	References
ZnOS Affinity	<i>eV</i>	4.10	4.30	[94, 95]
MoS ₂ Affinity	<i>eV</i>	4.00	4.25	
MoS ₂ thickness	<i>nm</i>	100	800	[44, 45, 67, 68]
MoS ₂ acceptor density	<i>cm⁻³</i>	1.0E+13	5.0E+15	
MoS ₂ donor density	<i>cm⁻³</i>	1.0E+12	5.0E+15	

Analysis of Variance (ANOVA), a statistical analysis tool, is used to test and validate the performance of selected parameters [97]. The ANOVA analysis results are given in Table 6.

Tablo 6.*PCE* function of SMZF solar cell based on ANOVA results.

Factor	Sum of Square	Df	Mean Square	F-Value	p-Value
A	8.8504	1	8.8504	75.576	0.000000
A²	3.6316	1	3.6316	31.011	0.000009
B	13.0086	1	13.0086	111.084	0.000000
B²	1.1539	1	1.1539	9.854	0.004313
C	458.9183	1	458.9183	3918.823	0.000000
C²	201.3239	1	201.3239	1719.157	0.000000
D	2.5268	1	2.5268	21.577	0.000093
D²	0.0832	1	0.0832	0.710	0.407277
E	103.2025	1	103.2025	881.273	0.000000
E²	35.2778	1	35.2778	301.246	0.000000
AB	3.2993	1	3.2993	28.174	0.000017
AC	1.5421	1	1.5421	13.168	0.001276
AD	0.0275	1	0.0275	0.235	0.631928
AE	6.9167	1	6.9167	59.063	0.000000
BC	0.5319	1	0.5319	4.542	0.043080
BD	0.4718	1	0.4718	4.029	0.055645
BE	16.5127	1	16.5127	141.006	0.000000
CD	2.0579	1	2.0579	17.573	0.000302
CE	11.7621	1	11.7621	100.439	0.000000
DE	0.0292	1	0.0292	0.249	0.621986
Error	2.9277	25			
MS Residual			0.1171		
Total SS	940.9962	45			
Predicted R²			0.994		
Adjusted R²			0.996		

Eq. (13) shows the actual regression equation obtained using Statistica v21. Here, A: ZnOS affinity, B: MoS₂ affinity, C: MoS₂ thickness, D: MoS₂ acceptor density, and E: MoS₂ defect density.

$$\begin{aligned}
PCE (\%) = & -323.13 + 209.414*A - 63.1384*A*A - 49.914*B - 26.5535*B*B + (13) \\
& 0.0171629*C - 0.0000390355*C*C + 2.90228E-15*D - 8.32836E- \\
& 33*D*D - 4.65181E-16*E + 3.05819E-31*E*E + 70.7003*A*B + \\
& 0.0176154*A*C + 3.36163E-16*A*D + 5.21064E-15*A*E - \\
& 0.00917803*B*C - 1.09994E-15*B*D - 5.60005E-15*B*E + 8.59092E- \\
& 19*C*D - 1.99183E-18*C*E + 1.36997E-32*D*E
\end{aligned}$$

The model can explain the *PCE* variation with a precision higher than 99 %. Table 6 shows not only the statistics but also how each physical parameter controls *PCE*. According to Table 6, the F-value indicates that the parameters with the highest physical influence/criticality are C, C², E, E², B, A, and D. This result shows that *PCE* is primarily controlled by the thickness of MoS₂ and the acceptor density. Furthermore, the model indicates that the combination of parameters D², AD, and DE are statistically insignificant terms. Furthermore, strong CE and BE interactions demonstrate that the MoS₂ layer acts as an active transport regulatory medium rather than a passive interface layer.

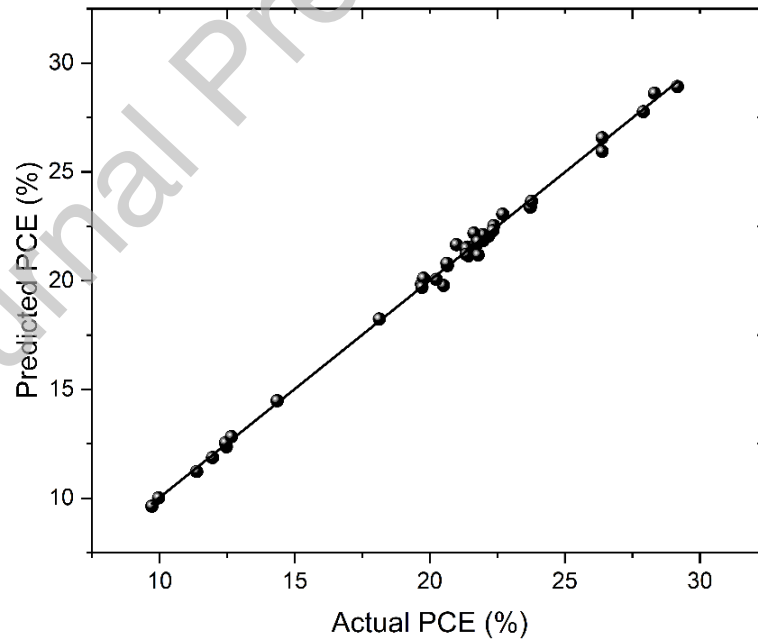


Fig. 5. Actual and predicted values of SMZF solar cell.

Fig. 5 clearly shows that there is a small difference between the actual and the predicted *PCE*. This strong agreement confirms the robustness of the optimization model for reliable device performance estimation.

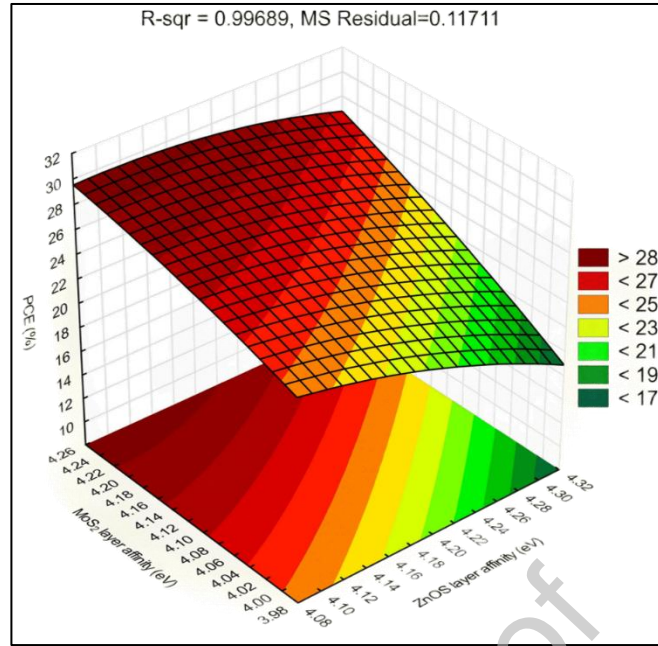


Fig. 6. The interaction effect of MoS₂ layer affinity and ZnOS layer affinity on SMZF solar cell using 3D Surface diagnosis in Statistica v21 software.

The ANOVA results confirm that the *PCE* exhibits a significant nonlinear dependence on the electron affinities of ZnOS and MoS₂, as also seen in the three-dimensional response surface model (Fig. 6). The parabolic curvature shows second-order effects, while the surface slope shows strong parameter interaction, consistent with the ANOVA findings. The high-efficiency plateau corresponds to the region determined by the GA, where band alignment is optimized, carrier transport is improved, and recombination is effectively suppressed [98]. The interaction effect between the MoS₂ layer defect density and MoS₂ layer electron affinity on SMZF solar cell, obtained from 3D Surface analysis in Statistica v21 software, is shown in Fig. 7. A gradual increase in the slope in one direction indicates a strong positive relationship between the variables.

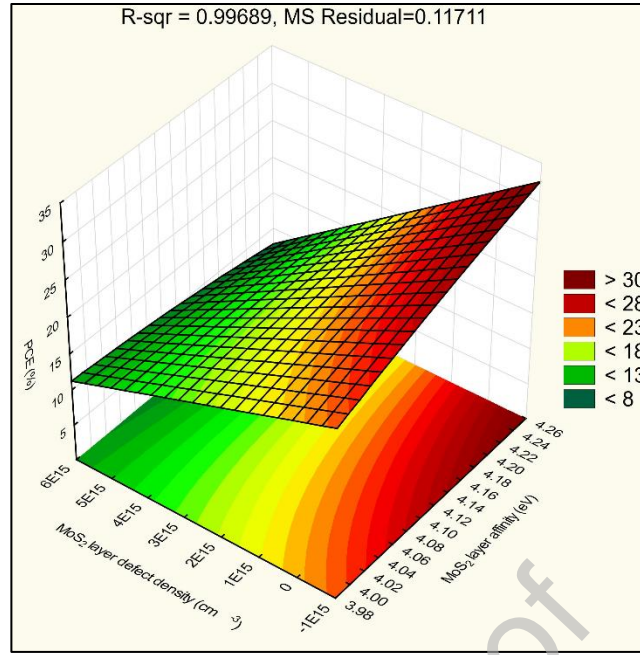


Fig. 7. The interaction effect of MoS₂ layer defect density and MoS₂ layer affinity on SMZF solar cell using 3D Surface diagnosis in Statistica v21 software.

3.4. Genetic Algorithm (GA)

The GA optimization tool is based on the principle of survival of the fittest [98]. Compared to classical approaches, GA enables the simultaneous optimization of structural and electronic parameters affecting device performance by effectively avoiding local optima through evolutionary processes such as selection, crossover, and mutation. In the present study, the fitness function used in the GA is the regression equation obtained from ANOVA. The GA optimization tool provides a delicate balance among band alignment, recombination, and carrier transport in photovoltaic structures. The flowchart of the implemented procedure is shown in Fig. 8.

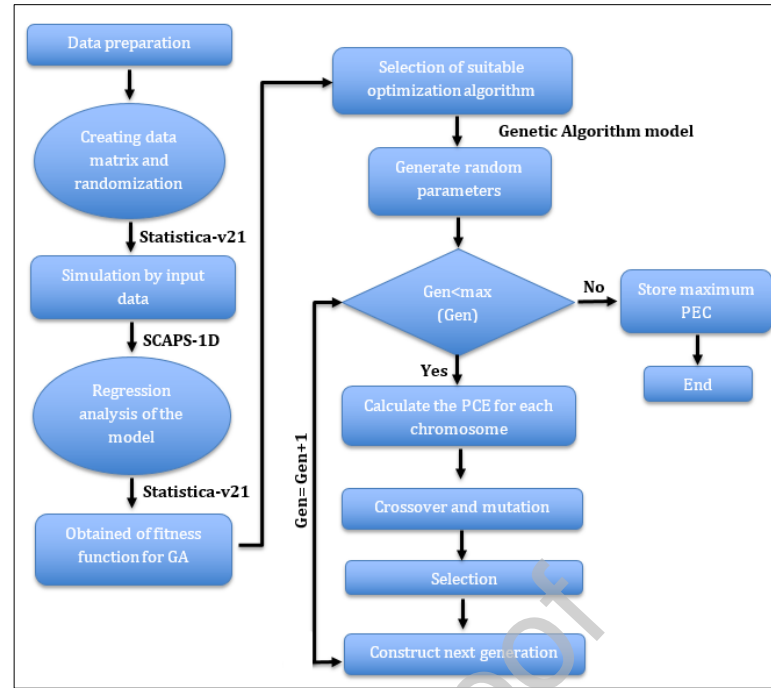


Fig. 8. Flow process chart used to optimize *PCE* of SMZF solar cell using RSM and GA.

The optimization of PCE by using a GA can be expressed as:

$$4.10 \text{ eV} < x(1) < 4.3 \text{ eV}$$

$$4.0 \text{ eV} < x(2) < 4.25 \text{ eV}$$

$$100 \text{ nm} < x(3) < 800 \text{ nm}$$

$$1.0\text{E}+13 \text{ cm}^{-3} < x(4) < 5.0\text{E}+15 \text{ cm}^{-3}$$

$$1.0\text{E}+12 \text{ cm}^{-3} < x(5) < 5.0\text{E}+15 \text{ cm}^{-3}$$

Here, $x(1)$: ZnOS Affinity, $x(2)$: MoS₂ Affinity, $x(3)$: MoS₂ Thickness, $x(4)$: MoS₂ acceptor density and $x(5)$: MoS₂ defect density are abbreviated. Table 7 also shows the GA parameters of the optimization parameter set of the SMZF solar cell structure.

Table 7.

GA parameters used for optimization of SMZF solar cell.

Parameters	Values
Population size	100
Scaling function	Rank
Function	Stochastic uniform
Mutation function	Adaptive feasible
Crossover function	Scattered
Generations	1000

The genetic algorithm optimization was performed in the MATLAB environment using real-coded individuals [99,100]. The initial population was randomly generated, and each individual represented a combination of variables within the predefined parameter boundaries. The objective function was defined as the power conversion efficiency (PCE) obtained from SCAPS-1D simulations. In the algorithm, a population size of 50 individuals, a crossover rate of 80%, and a mutation rate of 5% were employed. Elitism was applied to preserve the best solutions, while tournament selection was used during the selection process. The maximum number of generations was set to 100, and the algorithm was also terminated if the improvement became negligible over the last 20 generations. By utilizing a reduced search space previously defined through DOE–RSM, the computational burden was significantly decreased, enabling faster convergence [101].

Table 8 presents the best results and the optimum parameter set obtained using MATLAB's GA optimization tool, revealing that device performance is primarily governed by interface band alignment and recombination control. It can be interpreted that the optimized MoS₂ thickness (~723 nm) enhances the SMZF solar cell efficiency by providing sufficient optical absorption without increasing bulk recombination. Furthermore, maintaining the defect density at 10^{12} cm^{-3} may positively contribute to V_{oc} . All these parameters result in a balanced improvement in V_{oc} , FF , and PCE . Moreover, electron affinity values are identified as a dominant factor in the performance enhancement. Thus, it is concluded that GA identifies an optimal physical balance between PCE and the governing devices parameters for the proposed structure.

Table 8.

GA optimization results

Factors	Max. PCE (%)	ZnOS affinity (eV)	MoS ₂ affinity (eV)	MoS ₂ Thickness (nm)	MoS ₂ Acceptor density (cm^{-3})	MoS ₂ Defect Density (cm^{-3})
Results	31.416	4.1521	4.25	723.24	4.983E+15	1.0E+12

The fitness function is generally defined as negative of the objective function in GA studies. In this case, a decrease in the fitness value indicates that the population is improving. In our study, the highest PCE was obtained is 31.416 %. As shown in Fig. 9(a), the fitness value decreases rapidly during the initial generations; after approximately 60 generations,

convergence toward the optimum value begins, and the mean and best fitness curves approach each other.

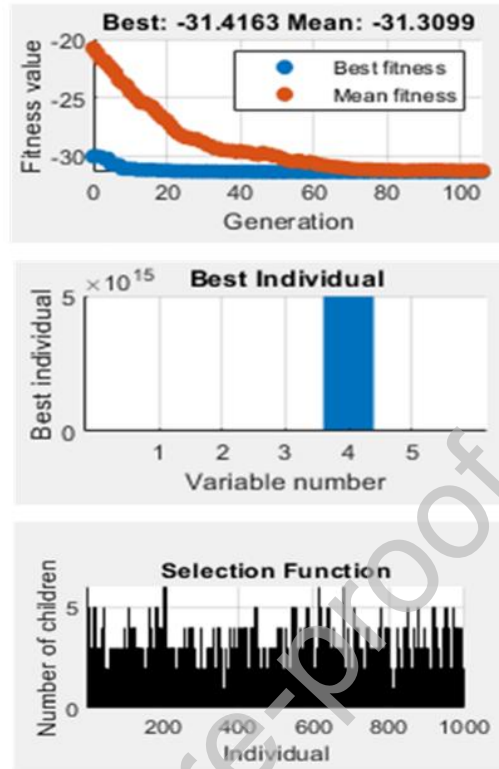


Fig. 9. The variation of fitness value across generation (a), the best individual performances (b), and selection function (c) of variables in coded form.

This result indicates that the generation is stable and that the optimization is reinforced across the entire population, rather than being driven by a single individual. As shown in Fig. 9 (b), the most dominant parameter in the GA is $x(4)$ (MoS_2 acceptor density). This result also demonstrates that acceptor doping concentration is one of the key factors determining device performance. Fig. 9 (c) shows that the population diversity is preserved and that the implemented algorithm is reliable. In conclusion, GA optimization not only increased PCE but also ensured a stable and reliable solution by effectively optimizing the device parameters.

3.5. Comparison with the Literature

The proposed solar cell structure differs from previously reported MoS_2 -based solar cells (Table 9) in three key aspects. First, whereas fixed bandgap electron transport layer (ETL) materials such as ZnO , CdS , or ZnSe are commonly used in the literature, this study employs ZnOS , an ETL material that has been extensively investigated in terms of band engineering.

The aim is not only material selection but also the precise control of the conduction band offset [69, 95, 98-100].

Tablo 9.

Compression of solar cells

Used analysis	Cell structure	J_{sc}	V_{oc}	FF	PCE (%)	References
AMPS-1D	n-MoS ₂ /p-Si	-	-	-	19.00	[45]
SCAPS-1D	n-MoS ₂ /p-MoS ₂	36.65	0.930	76.90	19.62	[42]
wxAMPS	ITO/ZnSe/MoS ₂ /SnS	29.89	0.841	85.00	21.39	[68]
SCAPS-1D	Al/FTO/CdS/MoS ₂ /Ni	34.11	0.760	83.00	21.61	[44]
AFORS-HET	Au/p-cSi/a-Si:H/n-MoS ₂ /graphene	44.25	0.697	82.95	25.61	[43]
AFORS-HET	TCO/n-type MoS ₂ /i-layer nc-Si:H/p-type c-Si/Al	42.81	0.771	82.42	27.22	[101]
SCAPS-1D	Al/FTO/CdS/p-MoS ₂ /Sb ₂ S ₃ /Ni	35.20	0.920	86.00	27.96	[44]
SCAPS-1D, BBD and GA	Au/Sb ₂ S ₃ /MoS ₂ /ZnOS/FTO	32.24	1.034	87.50	29.17	<i>This work, before optimization</i>
		34.30	1.039	88.16	31.42	<i>This work, after optimization,</i>

Second, the simultaneous improvement in V_{oc} (1.039 V), FF (88.16 %), and PCE (31.42 %) demonstrates that interface recombination is reduced and that device performance is consistently enhanced. Third, in contrast to traditional approaches in which only one parameter is varied while others are kept constant, this study employs a combined BBD-GA global optimization framework that captures the interdependencies among band alignment, doping density, and absorber thickness.

4. Conclusion

This study proposes a multivariate optimization strategy for SCAPS-1D-based solar cell design by integrating DOE-RSM and Genetic Algorithm (GA) methods, instead of the traditional OFAT approach. Using this framework, key device parameters such as absorber thickness, defect density, MoS₂/ZnOS electron affinity, and carrier concentration were simultaneously evaluated, and their interactions were statistically analyzed.

ANOVA and response surface analyses indicated that defect density and MoS₂ electron affinity are the most dominant parameters affecting power conversion efficiency (PCE). In addition, nonlinear and second-order interactions between the electron affinities of MoS₂ and ZnOS were identified.

The GA-based optimization significantly improved device performance, increasing the PCE from 29.17 % to 31.42 %. This enhancement was also reflected in improved V_{OC} , J_{SC} and FF values, and was validated by SCAPS re-simulation with a very low error (~0.013 %).

The optimized structure achieved a near spike/type-II band alignment at the MoS₂-ZnOS interface, which enhanced carrier collection efficiency and reduced interface recombination. The study demonstrates that the integrated DOE-GA approach provides a scalable and reproducible optimization framework not only for MoS₂-based systems but also for other thin-film and 2D-material-based photovoltaic architectures.

Additional information

Competing financial interests: The authors declare no competing financial interests.

CRedit authorship contribution statement

S. Çetinkaya and S. Yildirimcan: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Graphical abstract

